AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

DEPARTMENT OF ENGINEERING

INTRODUCTION TO DATABASE

SECTION: J

PROJECT NAME:

Procare Childcare Management System

Supervised By

KAWSER IROM RUSHEE

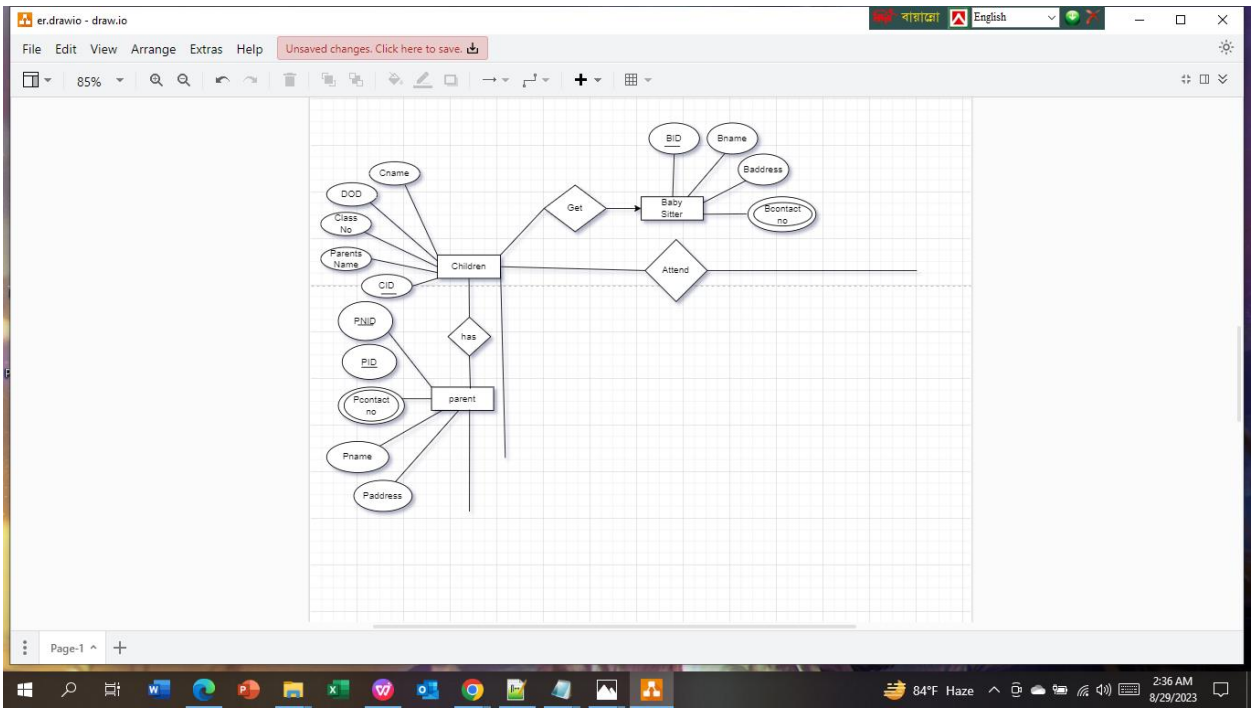
Submitted By

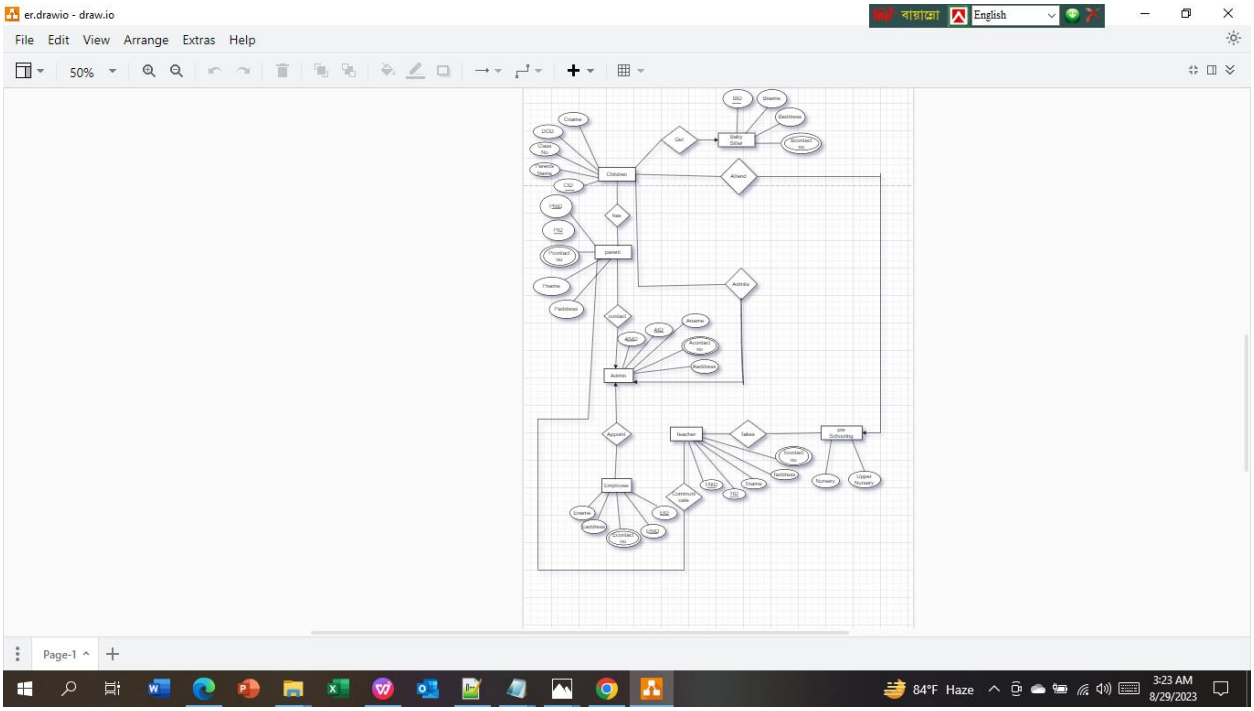
NAME	ID	GROUP NO
Saonty Saha	22-49206-3	9
Susmita Chakraborty Tanni	22-49208-3	9

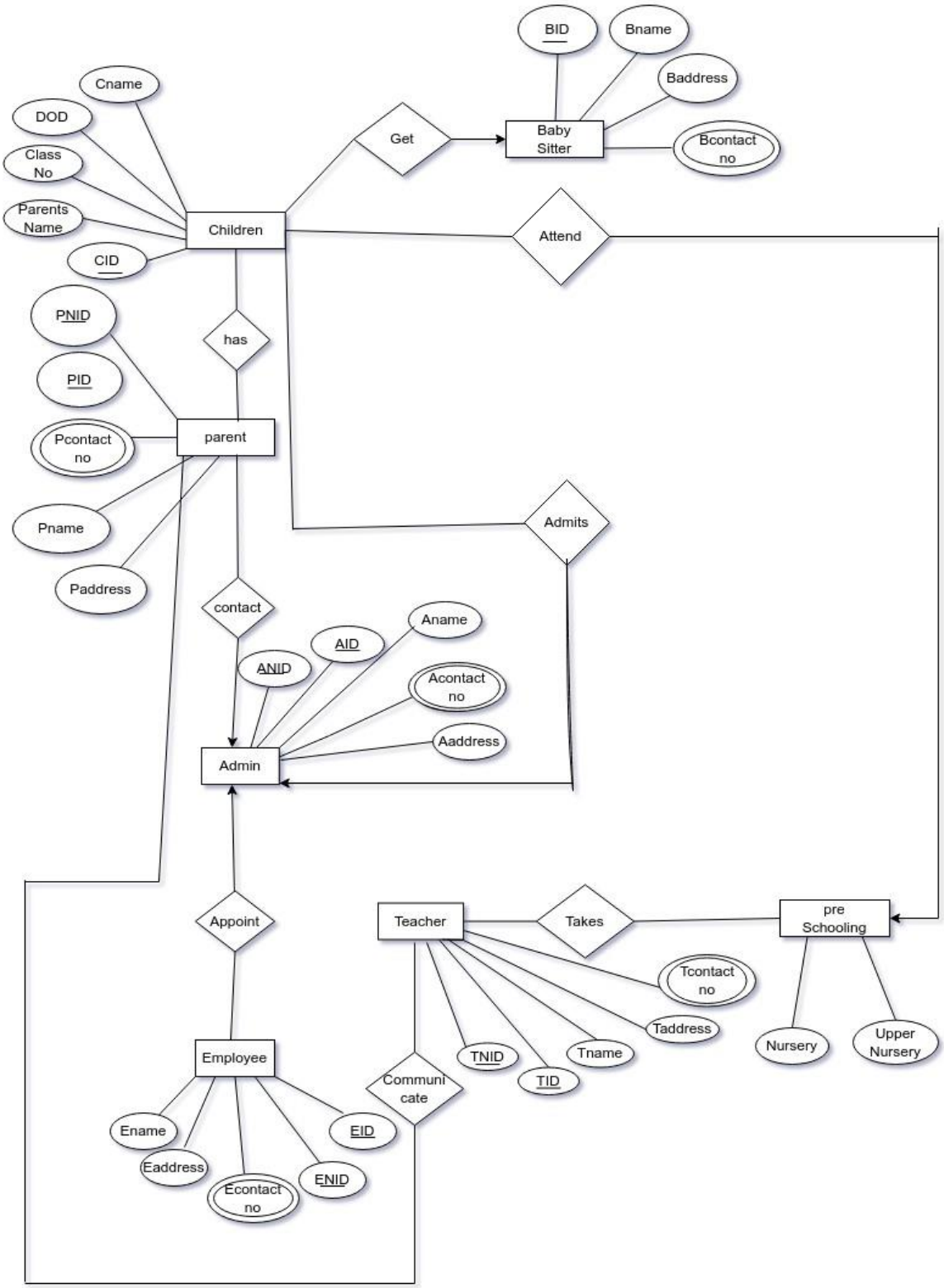
Case Study:

In Procare childcare management system, while parents are at work or are absent for another reason can keep their children.Parents can keep several children of them. Parents contact admin through Admin.Admin provides each parent with unique id and keep their address,several contact numbers, NID , name. admin. Admin appoints employees and provides distinctive id to employees. He keeps their name, address, NID,multivalued contact details of them to identify them. A teacher can take several classes but one class should have one teacher.Teacher's information like address,several contact numbers,unique ID,NID will be stored.Parents communicate with teachers for study progression purpose and so on. Children are admitted through admin regarding classes. One class can have a number of children but a child can be assigned to a particular class.This class is called pre schooling which is separated into nursery and upper nursey. Every children's name, id, class no., parents name, DOB will be stored.Every Children will get a babysitter.Every babysitters will have their name,unique id, address and several contact details. A babysitter can take care of abundant kids but a kid/child is provided with a babysitter .

ER Diagram:


er.drawio - draw.io
File Edit View Arrange Extras Help
Unsaved changes. Click here to save.
85%
BID
Bname
Baddress
Bcontact no
Baby Sitter
Get
Cname
DOD
Class No
Parents Name
CID
Children
Attend
PNID
PID
Pcontact no
Pname
Paddress
has
parent
Page-1
84°F Haze
2:36 AM
8/29/2023


er.drawio - draw.io
File Edit View Arrange Extras Help
50%
Page-1
84°F Haze
3:23 AM
8/29/2023


BID
Bname
Baddress
Bcontact no
Baby Sitter
Get
Cname
DOD
Class No
Parents Name
CID
Children
Attend
PNID
PID
has
Pcontact no
parent
Pname
Paddress
Admits
contact
Aname
AID
ANID
Acontact no
Aaddress
Admin
Appoint
Teacher
Takes
pre Schooling
Tcontact no
Taddress
Nursery
Upper Nursery
Employee
TNID
Tname
TID
Communi cate
Ename
Eaddress
Econtact no
ENID
EID

Normalization :

Get:

Unf: CID,Parents name,Class no,DOB,Cname,BID,Bname,Baddress,Bcontactno

1nf : CID,BID,Bcontactno,parents name,Class no,DOB,Cname,Bname,Baddress

2nf :1st:BID,CID

2nd:BID, Class No

3rd : CID , Cname,Class no,DOB,parents name

4th:BID,Bname,Baddress

3rd NF : 1st: BID, CID

2nd : BID, Class No

3rd : CID , Cname, Class no, DOB,Parents name

4th :CID, Class No

5th : BID, Bname,Baddress

Takes:

UNF: TNID, TID, Tname Taddress, T contactno,nursery,Upper Nursery

1stNF:TNID,TID,Tcontactno,Tname,Taddress,Nursery,

Upper nursery

2nf :1st : TID , Nursery, Upper Nursery

2nd : TNID, Tcontactno,Tname,Taddress

3nf : There is no transitive dependency

1st : TID, Nursery,Upper Nursery

2nd : TNID, Tcontactno,Tname,Taddress

Contact:

UNF : Pcontactno,Pname,Paddress,PID,PNID,PID,ANID,AID,

Aname,Acontactno,Aaddress

1nf : PNID,PID,Pcontactno,ANID,AID,Acontactno,Pname,

Padress, Aname,Aaddress

-2nf : 1st : PNID,PID,Pcontactno,Pname,Paddress

2nd : PNID,Pcontactno,Pname,Paddress

3rd :ANID,AID,Acontactno,Aname,Address

4th : ANID,,Acontactno,Aname,Address

3nf :There is no transitive dependency

1st : PNID,PID,Pcontactno,Pname,Paddress

2nd : PNID,Pcontactno,Pname,Paddress

3rd :ANID,AID,Acontactno,Aname,Address

4th : ANID,,Acontactno,Aname,Address

Communcate :

Unf :

PNID,PID,Pcontactno,Pcontactno,Pname,Paddress,TNID,TID,Tname,Taddr
ess,Tcontactno

1nf : PNID,PID,PContactno,TNID,TID,Tcontactno,
,Pname,Paddress,Tname,Taddress

2nf :

1st : PNID,PID,Pcontactno,Pname,Paddress

2nd : PNID,PID,Pcontactno,Pname,Paddress

3rd: TNID,TID,Tcontactno,Tname,Taddress

4th : TNID,Tcontactno,Tname,Taddress

3nf :There is no transitive dependency

1st : PNID,PID,Pcontactno,Pname,Paddress

2nd : PNID,PID,Pcontactno,Pname,Paddress

3rd: TNID,TID,Tcontactno,Tname,Taddress

4th : TNID,,Tcontactno,Tname,Taddress

Has:

UNF : CID,Parents name,Class no,DOB,Cname,Padress,Pname,Pcontactno,PID,PNID

1nf :

PNID,PID,Pcontactno,CID,Cname,Padress,Parentsname,Classno,DOB,Pname

2nf : 1st : PID,CID

2nd :PNID,Pcontactno,Padress,Pname,

3rd : CID,Cname,Classno,Parentsname,DOB

4th : CID, Classno

3nf : There is no transitive dependency

1st : PID,CID

2nd :PNID,Pcontactno,Padress,Pname,

3rd : CID,Cname,Classno,Parentsname,DOB

4th : CID, Classno

Admits:

UNF:Cname,DOD,Classno,Parentsname,CID,ANID,AID,Aname,Acontactno,Aadress,

1nf : CID,ANID,AID,Acontactno,Cname,DOB,Classno,Parentsname,Aname,Aadress

2nf : 1st : CID,DOB,Classno,Parentsname,

2nd : CID,Classno

3rd : ANID,AID,Acontactno,Aname,Aadress

4th : ANID,Acontactno,Aname,Aadress

3nf : There is no transitiove dependency

1st : CID,DOB,Classno,Parentsname,

2nd : CID,Classno

3rd : ANID,AID,Acontactno,Aname,Aadress

4th : ANID,Acontactno,Aname,Aadress

4th : ANID,Acontactno,Aname,Aadress

Attend:

Unf : CID,Parentsname,Classno,DOB,Cname,Nursery,Upper Nursery

1nf : Does not have any multivalued attribute

CID,Parentsname,Classno,DOD,Cname,Nursery,Upper Nursery

2nf : 1st : CID,Parentsname,Classno,DOB,Cname

2nd : Nursery,Upper Nursery

3rd : CID, Classno

3nf : There is no transitive dependency

1st:CID,Parentsname,Classno,DOB,Cname

2nd : Nursery,Upper Nursery

3rd : CID, Classno

Appoint:

UNF :

ANID,AID,Aname,Acontactno,Aadress,Ename,Eadress,Econtactno,ENID,EID

1nf :
ANID,AID,Acontactno,Econtactno,ENID,EID,Aname,Aadress,Ename,Eadr
ess

2nf: 1st : ANID,AID,Acontactno,Aname,Aadress

2nd : ANID, ,Acontactno,Aname,Aadress

3rd : ENID,EID,Econtactno,Ename,Eadress

4th : ENID,Econtactno,Ename,Eadress

3nf : There is no transitive dependency

1st : ANID,AID,Acontactno,Aname,Aadress

2nd : ANID, ,Acontactno,Aname,Aadress

3rd : ENID,EID,Econtactno,Ename,Eadress

4th : ENID,Econtactno,Ename,Eadress

Final Table: 1st: BID, CID

2nd : BID, Class No

3rd : CID , Cname, Class no, DOB,Parents name

4th :CID, Class No

5th : BID, Bname,Baddress

6th: TID, Nursery,Upper Nursery

7th :TNID, Tcontactno,Tname,Taddress

: 8Th: PNID,PID,Pcontactno,Pname,Paddress

9th : PNID,Pcontactno,Pname,Paddress

10th :ANID,AID,Acontactno,Aname,Address

11th : ANID,,Acontactno,Aname,Address

12th :TNID,TID,Tcontactno,Tname,Taddress

13Th : TNID,,Tcontactno,Tname,Taddress

14th : PID,CID

15th : Nursery,Upper Nursery

16th: ENID,EID,Econtactno,Ename,Eaddress

17th: ENID,Econtactno,Ename,Eaddress

Table Creation:

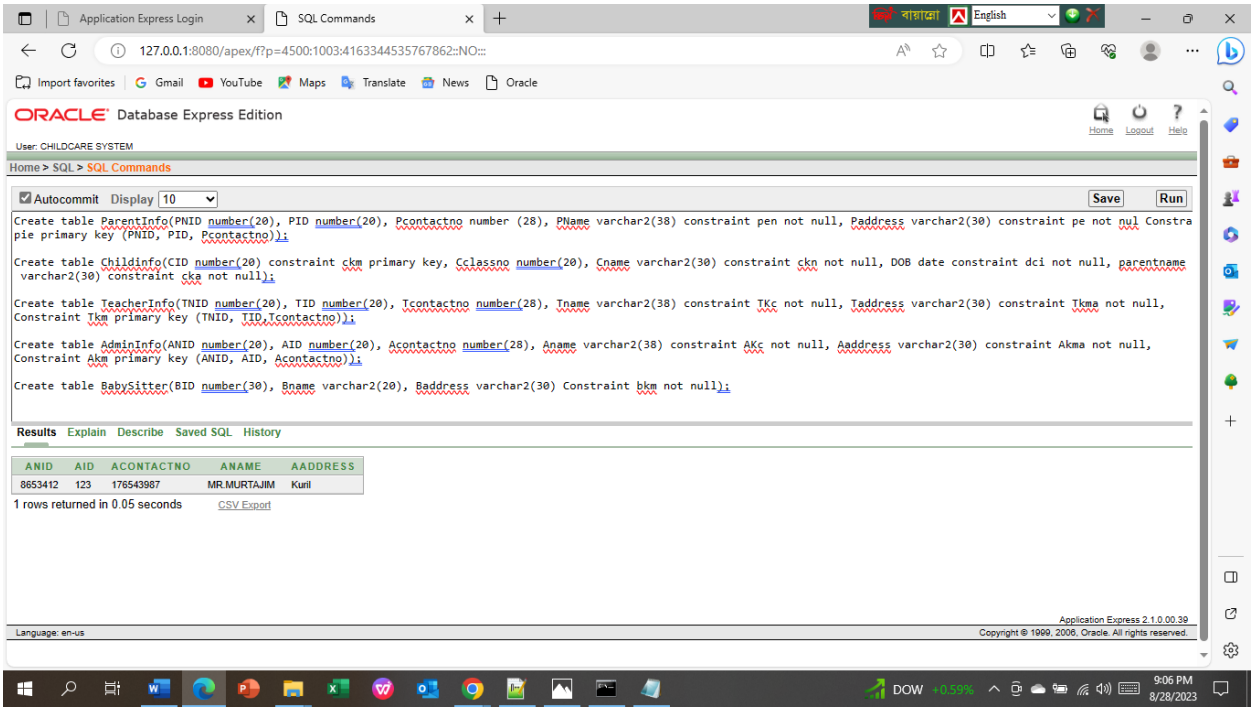
1.Create table ParentInfo(PNID number(20), PID number(20), Pcontactno number (28), PName varchar2(38) constraint pen not null, Paddress varchar2(30) constraint pe not nul Constraint pie primary key (PNID, PID, Pcontactno));

2.Create table Childinfo(CID number(20) constraint ckm primary key, Cclassno number(20), Cname varchar2(30) constraint ckn not null, DOB date constraint dci not null, parentname
varchar2(30) constraint cka not null);

3.Create table TeacherInfo(TNID number(20), TID number(20), Tcontactno number(28), Tname varchar2(38) constraint TKc not null, Taddress varchar2(30) constraint Tkma not null, Constraint Tkm primary key (TNID, TID,Tcontactno));

4.Create table AdminInfo(ANID number(20), AID number(20), Acontactno number(28), Aname varchar2(38) constraint AKc not null, Aaddress varchar2(30) constraint Akma not null, Constraint Akm primary key (ANID, AID, Acontactno));

5.Create table BabySitter(BID number(30), Bname varchar2(20), Baddress varchar2(30) Constraint bkm not null);



Joining:

1. select all the Babysitter information for the Ms. Monika (Equijoin)

```
Select B.*
From ChildInfo C,Babysitter B
Where C.CName=B.Bname and Parentname='Ms.Monika';
```

2. Get all the matching and non matching records from Babysitter and ChildInfo table.(Outer Join)

```
Select C.*,B.*
From ChildInfo C,Babysitter B
Where C.CName(+)=B.Bname;
```

3. Fetch all the parentInfo who are living Same address(Self join)

```
Select P.*,M.Pname,M.Paddress
From parentinfo P,parentinfo M
Where P.Paddress=M.Paddress and P.PID<>M.PID;
```

Subquery:

1.Display babysitter address whom does not live in Banani.

Select*

From Babysitter

Where Baddress in (select Baddress from Babysitter where Baddress not in 'Banani');

2. .Select all the parents name who lives in Kuril.

Select*

From parentinfo

Where Paddress in (select Paddress from parentinfo where Paddress ='Kuril');

3. Find teacher id where teacher id is greater than 4.

Select TID

From TeacherInfo

Where TID in (select TID from TeacherInfo where TID>4);